

Lecture 3: Thermodynamics

26021 Chemistry (Polytechnical Foundation)

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Introduction to thermodynamics

- ▶ *Thermodynamics, the scientific study of the interconversion of heat and other kinds of energy.*
- ▶ The laws of thermodynamics provide useful guidelines for understanding the energetics and directions of chemical reactions.
- ▶ We study changes in the state of a **system**, such as energy, temperature, pressure, and volume – all of which are **state functions** (determined by the state regardless of the path.)

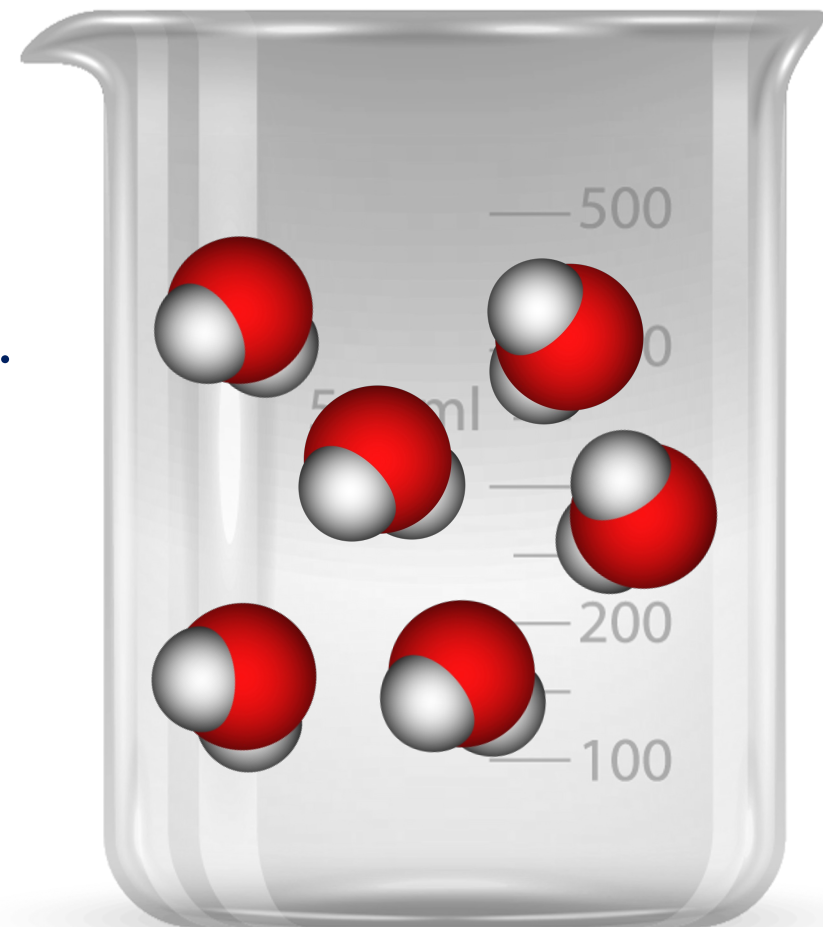
1st Law of Thermodynamics

Energy can be converted from one form to another, but cannot be created or destroyed.

What is the total internal energy of a system?

Kinetic energy: vibrations, rotations, movement of electrons,...

Potential energy: Attraction between electrons and nuclei, repulsion between electrons, interactions between molecules,...



1st Law of Thermodynamics

Energy changes, ΔU , are relevant and possible to quantify experimentally.

$$\Delta U = q + w \quad \rightarrow \text{Work exchange}$$

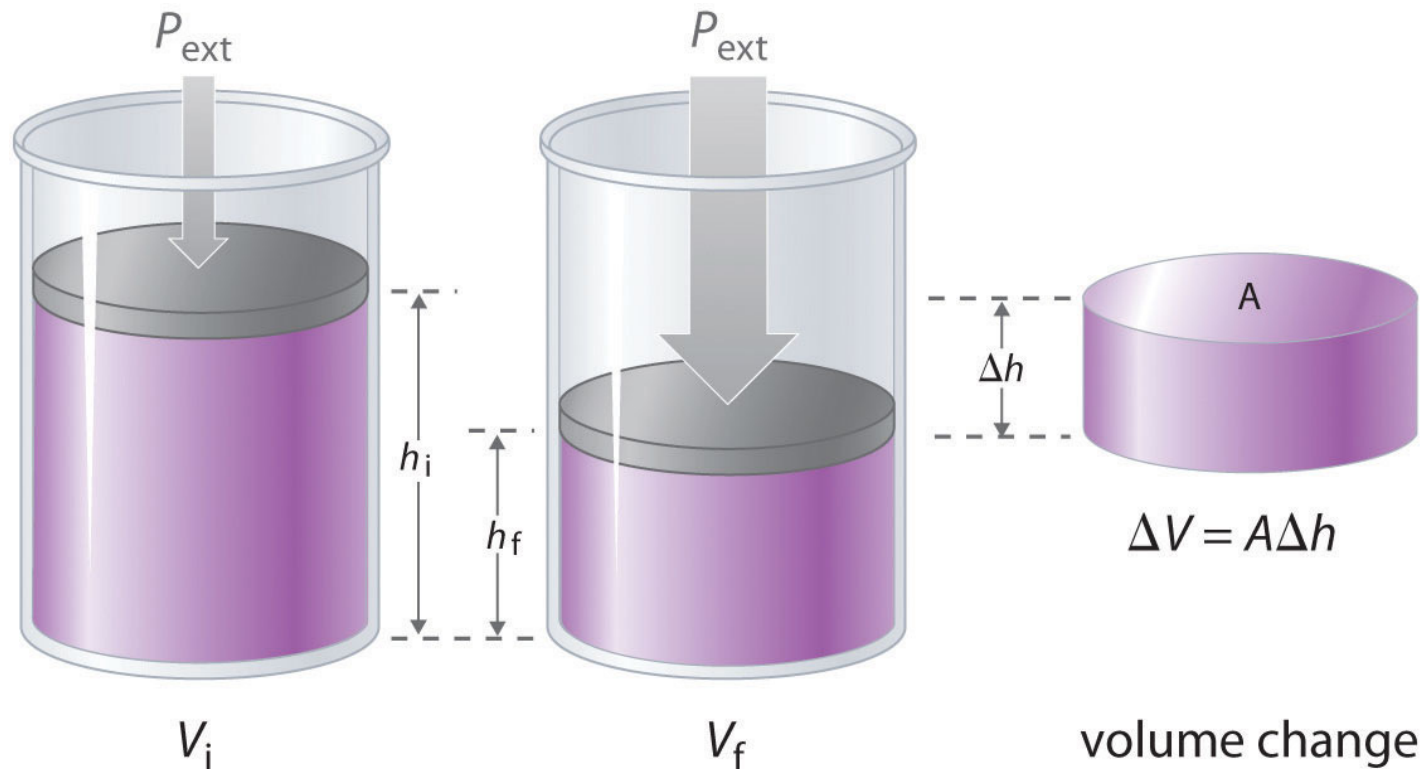
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Heat exchange

$q < 0$	heat released from the system to the surroundings: exothermic process
$q > 0$	heat absorbed by the system from the surroundings: endothermic process

Work, w

Defined (for now) as force, F , multiplied by distance, d

$$w = F \times d = F/d^2 \times d^3 = p \times V$$



Enthalpy

Most reactions occur under conditions of constant pressure, which means:

$$\Delta U = q + w = q_p - p\Delta V$$

Rearranging gives

$$q_p = \Delta U + p\Delta V = \Delta H$$

Here, $H = U + pV$, is the **enthalpy**.

For any given reaction, we can define an enthalpy of reaction, ΔH_{rxn} , as

$$\Delta H_{\text{rxn}} = H(\text{products}) - H(\text{reactants})$$

Enthalpy: examples

Melting of water:



Combustion of water:



Thermochemical equation guidelines:

- 1) Specify the physical state of all reactants and products
- 2) The enthalpy change reflects ΔH per mol of reaction.
- 3) If the reaction is reversed, the ΔH changes sign.

Standard Enthalpy

Our reference point is that the elements, in their most stable form at *standard conditions* (1 atm), by definition have a *standard enthalpy of formation*, ΔH_f° , of zero.

For example, for the reaction



Determine $\Delta H_f^\circ(\text{diamond})$?

$$\begin{aligned}\Delta H_{\text{rxn}}^\circ &= H^\circ(\text{products}) - H^\circ(\text{reactants}) \\ &= H_f^\circ(\text{graphite}) - H_f^\circ(\text{diamond}) \\ &= 0 \text{ kJ/mol} - H_f^\circ(\text{diamond}) = -1.90 \text{ kJ/mol}\end{aligned}$$

$$\Delta H_f^\circ(\text{diamond}) = \underline{1.90 \text{ kJ/mol}}$$

EXAMPLES

- 1) Dissolution of H_2SO_4 in water
- 2) Dissolution of NH_4NO_3 in water
- 3) $\text{KClO}_3(\text{s}) + \text{P}_4(\text{s}) \rightarrow \text{P}_4\text{O}_{10}(\text{s}) + \text{KCl}(\text{s})$

	<u>ΔH_f° (kJ mol⁻¹)</u>
$\text{KClO}_3(\text{s})$	-391
$\text{P}_4(\text{s})$	0
$\text{P}_4\text{O}_{10}(\text{s})$	-3010
$\text{KCl}(\text{s})$	-436

Redox reactions – how to balance?

Conservation of electrons, atoms, and charge

Procedure:

- (1) Identify half-cell reactions
- (2) Balance the electron count by a common denominator and add half-reactions
- (3) Balance charges using H^+ in acidic solution, OH^- in basic solution (or either in neutral solution). Finally, add H_2O to balance.

Hess's law

When reactants are converted to products, the change in enthalpy is the same whether the reaction takes place in one step or in a series of steps.

EXAMPLE

Combustion of graphite to form carbon monoxide

Spontaneous change

A ***spontaneous change*** is one that occurs ***without*** a continuous input of energy from outside the system.

All chemical processes require energy (***activation energy***) to take place, but once a spontaneous process has begun, no further input of energy is needed.

A ***nonspontaneous*** change occurs only if the surroundings ***continuously*** supply energy to the system.

If a change is spontaneous in one direction, it will be nonspontaneous in the reverse direction.

ΔH does not predict spontaneity

A spontaneous change may be exothermic or endothermic.

Spontaneous **exothermic** processes include:

- freezing and condensation at low temperatures,
- combustion reactions,
- oxidation of iron and other metals.

Spontaneous **endothermic** processes include:

- melting and vaporization at higher temperatures,

The sign of ΔH does not by itself predict the direction of a spontaneous change.

Entropy

The number of microstates (W) in a system is related to the **entropy** (S) of the system.

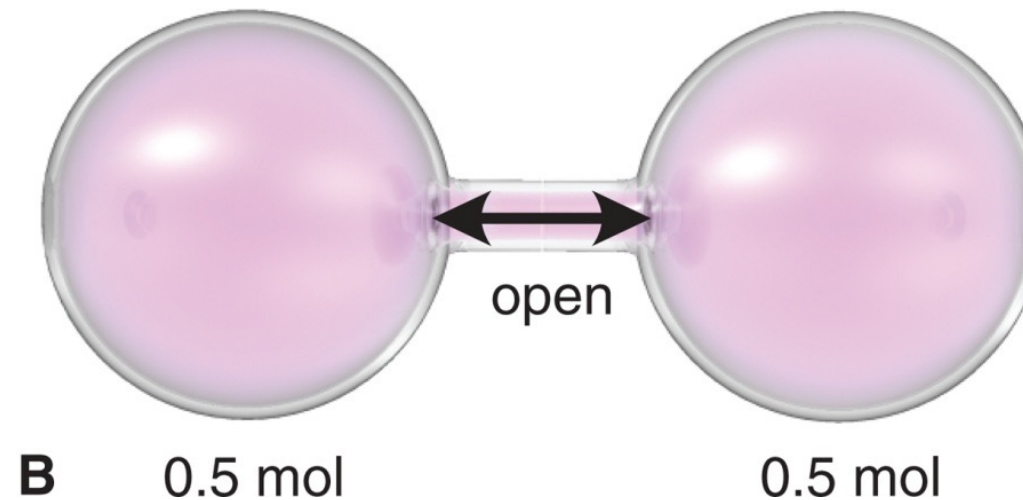
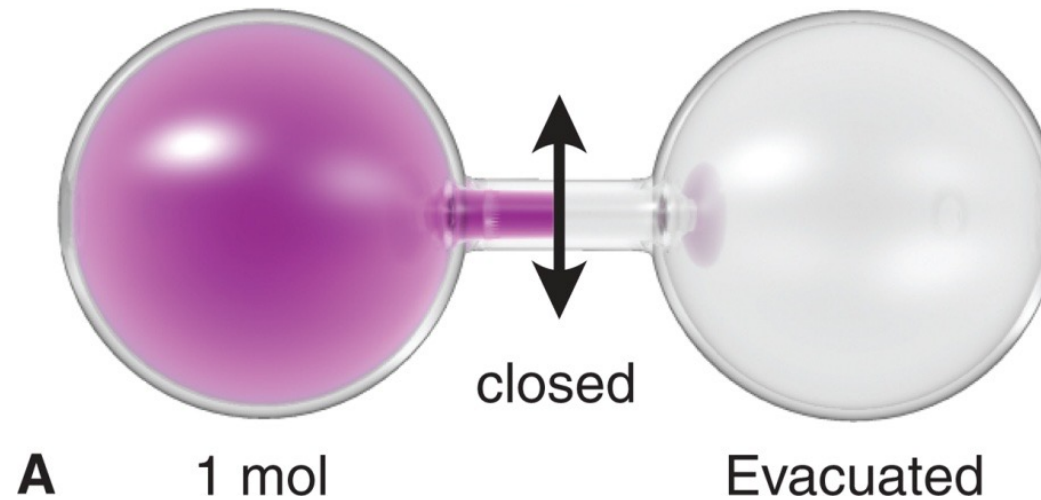
$$S = k_B \ln(W)$$

A system with ***fewer*** microstates has ***lower entropy***.

A system with ***more*** microstates has ***higher entropy***.

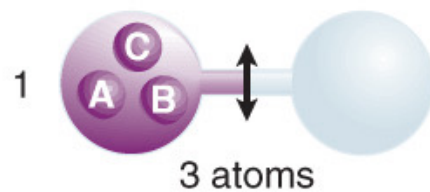
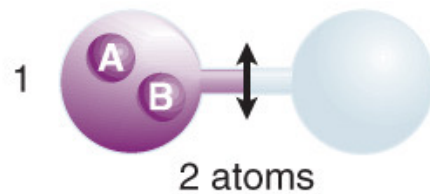
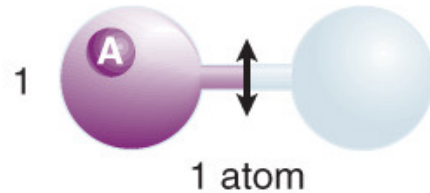
Entropy, like enthalpy, is a ***state function***

Example: Expansion of gas

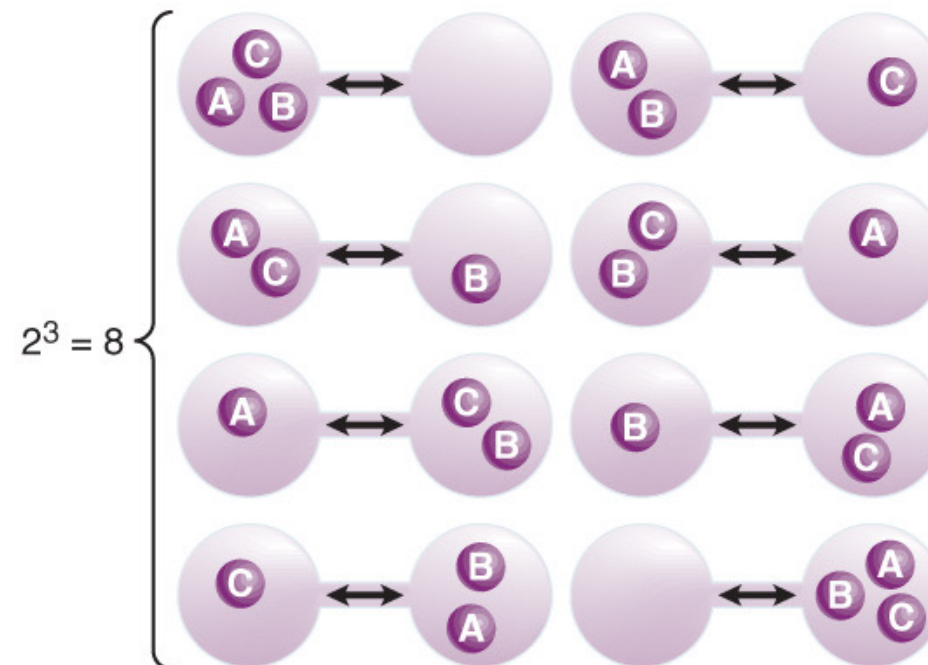
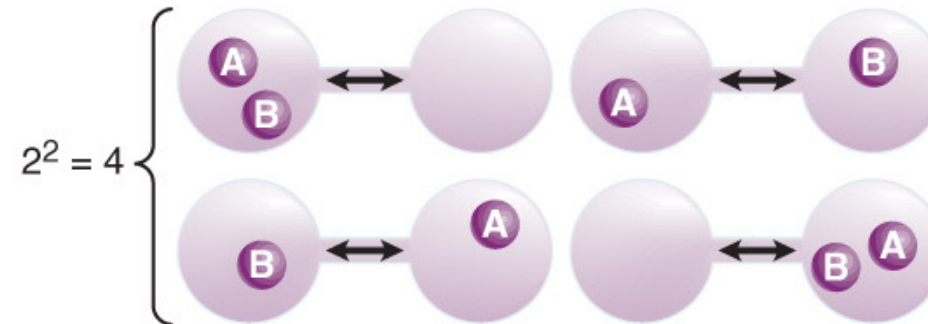
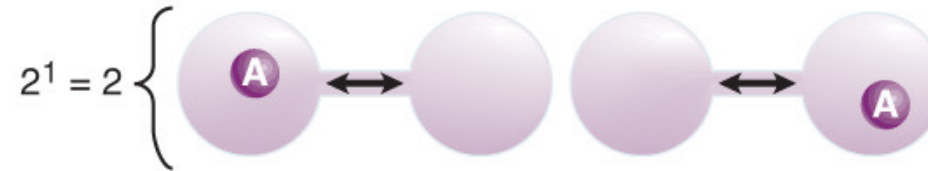


Example: Expansion of gas

W CLOSED

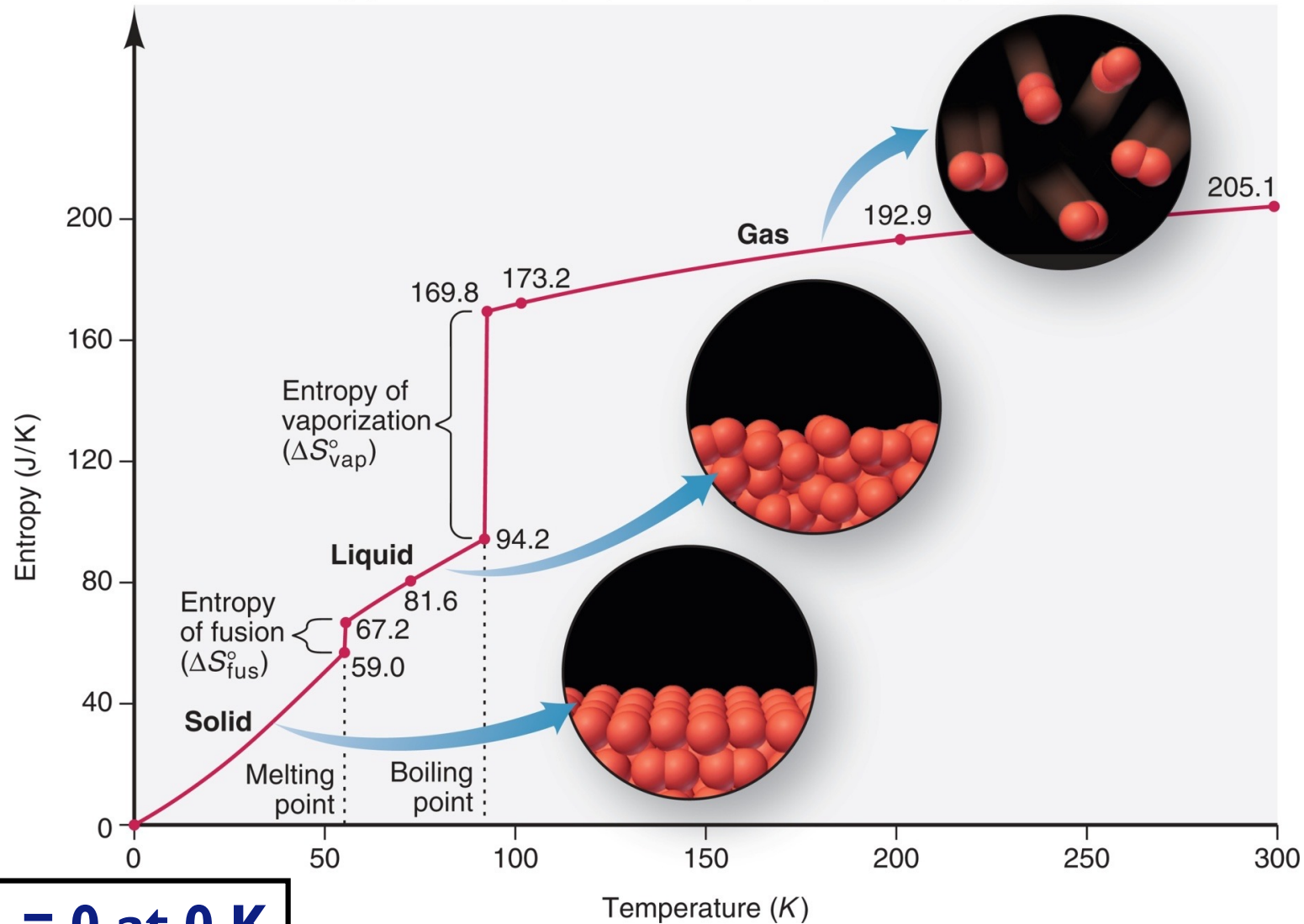


W OPEN



Entropy changes by temperature

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$$S_{sys} = 0 \text{ at } 0 \text{ K}$$

The 2nd Law of Thermodynamics

The entropy of the universe increases in a spontaneous process and remains unchanged in an equilibrium process.

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$$

Spontaneous process

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} = 0$$

Equilibrium process

The 2nd Law of Thermodynamics

The entropy of the universe increases in a spontaneous process and remains unchanged in an equilibrium process.

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} > 0$$

Spontaneous process

Change in ΔS_{surr} is proportional to ΔH_{sys}

$$\Delta S_{\text{surr}} \propto -\Delta H_{\text{sys}}$$

In other words, an exothermic reaction ($\Delta H_{\text{sys}} < 0$) will lead to $\Delta S_{\text{surr}} > 0$ (e.g. enhances motion of molecules in surroundings). This will have little effect if the temperature is already high, therefore

$$\Delta S_{\text{surr}} = -\Delta H_{\text{sys}}/T$$

Clausius theorem (1855)

ΔS and spontaneity

For any spontaneous process:

$$\begin{aligned}\Delta S_{\text{univ}} &= \Delta S_{\text{sys}} + \Delta S_{\text{surr}} \\ &= \Delta S_{\text{sys}} + [-\Delta H_{\text{sys}}/T] > 0\end{aligned}$$

which allows us to define the *Gibbs free energy* change:

$$\Delta G_{\text{sys}} = -T \Delta S_{\text{univ}} = \Delta H_{\text{sys}} - T \Delta S_{\text{sys}} < 0$$

(multiplying by $-T$ and reversing the inequality sign)

$\Delta G < 0$ for a spontaneous process

$\Delta G > 0$ for a nonspontaneous process

$\Delta G = 0$ for a process at equilibrium

Effect of T on Reaction Spontaneity

$$\Delta G = \Delta H - T \Delta S$$

Reaction is spontaneous at all temperatures

If $\Delta H < 0$ and $\Delta S > 0$

$\Delta G < 0$ for all T

Reaction is nonspontaneous at all temperatures

If $\Delta H > 0$ and $\Delta S < 0$

$\Delta G > 0$ for all T

Effect of T on Reaction Spontaneity

**Reaction becomes spontaneous as
 T increases**

If $\Delta H > 0$ and $\Delta S > 0$

ΔG becomes *more negative* as T
increases.

**Reaction becomes spontaneous as
 T decreases**

If $\Delta H < 0$ and $\Delta S < 0$

ΔG becomes *more negative* as T
decreases.

EXERCISE

Consider this reaction:



	ΔH_f° (kJ mol ⁻¹)	ΔS_f° (J K ⁻¹ mol ⁻¹)
NO₂	33.2	240
N₂O₄	9.16	304

Is the reaction to the right spontaneous at:

- a) 100 °C?
- b) -78 °C?

You can assume that the standard enthalpies and entropies do not change with temperature

EXERCISE

PROBLEM: A key step in the production of sulfuric acid is the oxidation of $\text{SO}_2(\text{g})$ to $\text{SO}_3(\text{g})$:



At 298 K, $\Delta G = -141.6 \text{ kJ/mol}$; $\Delta H = -198.4 \text{ kJ/mol}$; and $\Delta S = -187.9 \text{ J/K}$

- (a) Use the data to decide if this reaction is spontaneous at 25 deg. C, and predict how ΔG will change with increasing T .
- (b) Assuming ΔH and ΔS are constant with increasing T , is the reaction spontaneous at 900 deg. C?